

Algebraic Cobordisms

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1 Crash Course Through Topological Cobordisms and Spectra

We consider the following topological problem. Given a closed manifold M , when does it "bound"; i.e. is the boundary of a compact manifold? To answer this, we propose the following moduli problem:

Definition 1. Given closed manifolds M, N , they are said to be **bordant** if the disjoint union $M \sqcup N$ bounds. Then the bounding manifold (namely W where $\partial W = M \sqcup N$) is said to be a cobordism between M, N . The orientable manifolds M, N are said to be **oriented bordant** by W if $\partial W = M \sqcup (-N)$.

Remark 1. One can readily see that bordism is an equivalence relation on the category of (smooth/topological) manifolds. Therefore, we get equivalence classes Ω_k of k -dimensional manifolds. It is then immediate that each bordism class is a group via the operation of disjoint union. Note that this is an Abelian group and a vector space over $\mathbb{F}_2$. Therefore, we refer to this as the **bordism group** of k -dimensional manifolds. One can form the commutative graded ring $\Omega_* = \bigoplus \Omega_k$ (note that this is not graded commutative) where the multiplication is defined as the product of manifolds. We refer to this as the **bordism ring** of smooth/topological manifolds.

The initial problem is then to identify the identity element of the bordism groups.

Exercise 1. If two manifolds M, N bound higher-dimensional compact manifolds, so does their connected sum.

Lemma 1. If a closed surface Σ bounds; i.e. is the boundary of a compact 3-manifold, then $\chi(\Sigma) \stackrel{2}{=} 0$.

Proof. Let $\partial M = \Sigma$ for a 3-dimensional manifold M . Let $2M = M \sqcup M/\partial M$. Since this is a closed, odd-dimensional manifold without boundary, $\chi(2M) = 0$. However, we have by using the Mayer-Vietoris sequence that $\chi(2M) = 2\chi(M) - \chi(\partial M)$, which completes the proof of the lemma. $\square$

Exercise 2. Using classifications of closed 0,1,2 dimensional manifolds, prove that $\Omega_0 = \mathbb{Z}_2, \Omega_1 = 0, \Omega_2 = \mathbb{Z}_2$.

Reminder. A **spectrum** E is a collection of pointed CW-complexes $\{E(n)\}_{n \in \mathbb{Z}}$ equipped with maps of pointed spaces $S^1 \wedge E(n) \rightarrow E(n+1)$. Given a pointed finite CW-complex X , denote by $[X, E(n)]$ the homotopy classes of maps of pointed spaces $X \rightarrow E(n)$. For any $f \in [X, E(n)]$, there is a composition $SX := S^1 \wedge X \xrightarrow{S \wedge f} S^1 \wedge E(n) \rightarrow E(n+1)$ which we denote by Sf . Therefore, we get a direct system of homotopy groups $[S^k X, E(n+k)]$. We say that E is **convergent** if each $E(n)$ is $(n-1)$ -connected; i.e. $\pi_i(E(n)) = 0, \forall i < n$.

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Rokhlin proved (via techniques of surgery theory) that $\Omega_3 = 0$. The calculation of higher cobordism groups were hopeless until Thom's seminal result. Let $BU(n)$ be the moduli space of all rank n vector bundles over X , and let γ^n be the universal n -bundle. Then every rank n vector bundle over X is the pullback of γ^n under some map $X \rightarrow BU(n)$. We then take the Thom space $T(\gamma^n)$, given by taking the quotient of the sphere bundle $S(\gamma^n)/X$. This means to take one-point compactification of each fibre and then identify all the new points as a single point, being the basepoint of $T(\gamma^n)$. By the canonical inclusions $BU(n) \hookrightarrow BU(n+1)$ (given by adding a trivial line bundle), we get a spectrum over X which we call the **Thom spectrum** MU , where $MU(n) = T(\gamma^n)$.

Definition 2. A **function of spectra** $f : E \rightarrow F$ is a collection of cellular maps $\{f_n : E(n) \rightarrow F(n)\}_{n \in \mathbb{Z}}$ such that $f_{n+1} \mid_{SE(n)} = S f_n$ for all integers n . Given a spectrum M , a **subspectrum of M** is a spectrum M' such that $M'(n)$ is a subcomplex of $M(n)$ for all integers n . The inclusion $i : M' \hookrightarrow M$ is then a function of spectra.

Definition 3. Given a spectrum E , if e_n^d is a d -cell of the cell complex $E(n)$, then Se_n^d is a $(d+1)$ -cell of $E(n+1)$. We say that a cell e_n^d is **primitive** if it is not the suspension of a $(d-1)$ -cell of the cell complex $E(n-1)$. Given a primitive cell e_n^d , the sequence $e = \{e_n^d, Se_n^d, S^2e_n^d, \dots\}$ is called a **cell of dimension $d-n$** in E . Note that each cell of a cell complex $E(i)$ is a member of exactly one cell of E .

Definition 4. A spectrum E is **finite/countable** if it has only finitely/countably many cells.

Exercise 3. E is finite if and only if there exists some finite $E(N)$ such that $E(n) = S^{n-N}E(N)$, $\forall n \geq N$.

Definition 5. A subspectrum $F \subseteq E$ is said to be **cofinal** if each cell of E ultimately lies in F ; that is, for any cell $e_n^d \subseteq E(n)$, there exists some integer m such that $S^m e_n^d \subseteq F(n+m)$.

Remark 2. If F is cofinal and $K \subseteq E(n)$ is a finite subcomplex, there exists $m \in \mathbb{Z}$ such that $S^m K \subseteq F(n+m)$.

Definition 6. Let E, F be spectra. We call a function of spectra $f' : E' \rightarrow F$ a **local function** if $E' \subseteq E$ is a cofinal subspectrum. We say $(f' : E' \rightarrow F) \sim (f'' : E'' \rightarrow F)$ if they agree on a cofinal common subspectrum; that is, there exists a local function $f''' : E''' \rightarrow F$ and $E''' \subseteq E' \cap E''$ is a cofinal subspectrum and $f' \mid_{E'''} \cong f'' \mid_{E'''}$. This is an equivalence relation. We call each equivalence class a **map of spectra**.

Exercise 4. Let $f : E \rightarrow F$ be a function of spectra. If $F' \subseteq F$ is a cofinal subspectrum, there is a cofinal subspectrum $E' \subseteq E$ such that $f(E') \subseteq F'$. Note that this gives a well-defined composition of maps of spectra.

Remark 3. The intersection of two cofinal subspectra is cofinal. Arbitrary union of cofinal subspectra is cofinal. If $G \subseteq F$ is cofinal and $F \subseteq E$ is cofinal, then $G \subseteq E$ is cofinal. The inclusion $i : F \hookrightarrow E$ of a subspectrum defines a map of spectra. Given any map of spectra $g : E \rightarrow G$, the restriction $g \mid_F := g \circ i$ is a map of spectra.

Definition 7. Given spectra E, F , we define a **homotopy of spectra** as a map of spectra $h : E \wedge I \rightarrow F$. The inclusions $0, 1 \in I$ induce maps of spectra $i_0, i_1 : E \rightarrow E \wedge I$. We say two maps of spectra $f_0, f_1 : E \rightarrow F$ are **homotopic** if there is a homotopy $h : E \wedge I \rightarrow F$ such that $h_0 := h \circ i_0 \cong f_0$, $h_1 := h \circ i_1 \cong f_1$.

We denote by $[E, F]$ the set of homotopy classes of maps of spectra $E \rightarrow F$. Composition of maps passes to $[E, F]$.

In terms of cofinal subspectra, two maps $f_0, f_1 : E \rightarrow F$ represented by local functions $f_0' : E_0' \rightarrow F, f_1' : E_1' \rightarrow F$ respectively are homotopic if there is a cofinal common subspectrum $E'' \subseteq E_0' \cap E_1'$ and a function of spectra $h'' : E'' \wedge I \rightarrow F$ such that $h_0'' := h'' \circ i_0 \cong f_0' |_{E''}, h_1'' := h'' \circ i_1 \cong f_1' |_{E''}$

Example 1.

1. Given a CW-complex X , the **spectrum $E(X)$** is defined by $E(X)(n) = S^n X, E(X)(-n) = *, \forall n \geq 0$. We denote by $S^* = E(S^0)$ the **sphere spectrum**; that is, $S^*(n) = S^n$.
2. Given a spectrum E and a pointed CW-complex (X, x_0) , we can form the spectrum **spectrum $E \wedge X$** given by $(E \wedge X)(n) = E(n) \wedge X$ equipped with the weak topology. Note that $S(E \wedge X)(n) = S(E(n) \wedge X) = S^1 \wedge (E(n) \wedge X) \cong (S^1 \wedge E(n)) \wedge X \rightarrow E(n+1) \wedge X$, which gives $E \wedge X$ the structure of a spectrum.
3. Given a map $f : E \rightarrow F$ of spectra, we define $CE = E \wedge I$ and construct the **mapping cone $F \cup_f CE$** as defined by $(F \cup_f CE)(n) = F(n) \cup_{f'_n} (E'(n) \wedge I)$ where the function $f' : E' \rightarrow F$ represents the map f .
4. Given a spectrum E , we can define the **spectrum ΣE** as $\Sigma E(n) = E(n+1), \forall n \in \mathbb{Z}$.
5. Given a collection $\{E_i\}_{i \in I}$ of spectra, we define the **wedge spectrum $\bigvee_i E_i$** by $(\bigvee_i E_i)(n) = \bigvee_i E_i(n)$. This is a spectrum given due to the maps $S(\bigvee_i E_i(n)) = \bigvee_i S E_i(n) \rightarrow \bigvee_i E_i(n+1)$

In the early of the 1950's Eilenberg–Steenrod discovered that the homology theory $H_*(-, G)$ as a functor on the category of finite CW-complexes is determined by certain axioms, called thereafter the Eilenberg–Steenrod axioms. Later (end of the 50's, beginning of the 60's) it was noticed that many useful constructions of algebraic topology (K-functor, bordism groups, etc.) are formally similar to (co)homology theories. Afterwards the reason for this phenomenon was clarified: namely, most of these constructions satisfy all the Eilenberg–Steenrod axioms except the so-called dimension axiom. So, it seemed reasonable to consider the objects satisfying these axioms. These objects were called extraordinary (co)homology theories. However, later mathematicians came to call these objects just (co)homology theories, while $H(-, G)$ got the name ordinary (co)homology theory.¹

Definition 8. Given a spectrum E and a pointed CW-complex (X, x_0) , we can define $E_n(X) = \pi_n(E \wedge X) = [\Sigma^n S^*, E \wedge X]$ and $E^n(X) = [E(X), \Sigma^n E] \cong [\Sigma^{-n} S^* \wedge X, E]$. By the canonical prism maps, one can show that these give (reduced) homology and cohomology theories.

Thom proved that the cobordism groups are stable homotopy groups of the Thom spectrum: $\Omega_n = \varinjlim \pi_{n+k} MU(k)$, which will be shown to be the same as $MU_n(\text{pt})$. In fact, the oriented bordism groups are also given by $\varinjlim \pi_{n+k} MSU(k)$, where MSU is constructed similar to MU , but instead beginning with the universal oriented vector bundles of each rank. To study the cobordism ring, we define the relative moduli problem which is relative to an arbitrary manifold.

Definition 9. Given a manifold X and homeomorphisms of manifolds $f : M \rightarrow X, g : N \rightarrow X$, the pairs $(M, f), (N, g)$ are said to be **bordant over X** if there exists a pair (W, F) such that W is a bordism between M, N (i.e. $\partial W = M \sqcup N$) and $F|_M \cong f, F|_N \cong g$. Similarly, we get bordism groups of k -dimensional manifolds $\Omega_n(X)$ that are still Abelian. This gives functors $\text{Top} \rightarrow \text{AbGrp}$, which is homotopy-invariant. We also get the bordism ring over X given by $\Omega_*(X)$ that is commutative graded, which gives a functor $\text{Top} \rightarrow \text{GrRng}$.

¹In fact, the classical (co)homology should be called extraordinary, because it has a certain extraordinary property: it satisfies the dimension axiom. (This is note of Peter Hilton)

Example 2. The bordism groups $\Omega_n(\text{pt})$ of a point are the ordinary bordism groups Ω_n . In fact, $\Omega_*(X)$ is a module over $\Omega_*(\text{pt}) = \Omega_*$ for an arbitrary manifold X . The module structure is given by $[N].[M, f] = [N \times M, f \circ \pi_M]$, $[N] \in \Omega_*$, $[M, f] \in \Omega_*(X)$ where $\pi_M : N \times M \rightarrow M$ is the projection.

By Thom's seminal theorem, the bordism groups over X satisfy all properties of a generalized (co)homology theory, and therefore give fundamental algebraic invariants for study of manifolds. In fact, Quillen proved that this (co)homology theory is to some sense universal; that is, given any other complex-oriented (co)homology theory E^* (a cohomology theory that assigns chern classes to line bundles), admits a unique "oriented" natural transformation $MU^* \rightarrow E^*$. Following the promising and fruitful study of cobordisms in differential topology, we intend to create a replica of this theory in the setting of algebraic geometry. This lets us achieve an almost universal cohomology theory of schemes.

2 Structure Theorems of Cobordism Rings

Back to the original moduli problem of identifying (automatically closed) smooth manifolds that bound a compact smooth manifold, we review some structure theorems that classify such manifolds based on their algebraic invariants.

Oriented Cobordisms

Proposition 1. (Thom, Pontrjagin) A closed k -dimensional smooth manifold bounds (i.e. is the boundary of a compact $(k+1)$ -dimensional smooth manifold) if and only if its Stiefel-Whitney numbers are zero.

Reminder. One can define the Stiefel-Whitney classes of a vector bundle $\pi : \xi \rightarrow M$ as cohomology classes $\omega_i(\xi) \in H^i(B(\xi), \mathbb{Z}/2\mathbb{Z})$ (consult [1]). Therefore, we can define Stiefel-Whitney classes of a smooth manifold M to be the Stiefel-Whitney classes of its tangent bundle; that is, $\omega_i(M) := \omega_i(TM)$. Given positive integers $i_1 + \dots + i_n = k$, we can form the cup product $\omega_{i_1} \smile \dots \smile \omega_{i_n} \in H^k(M, \mathbb{Z}/2\mathbb{Z})$ (we will refer to this as $\omega_{i_1} \cdots \omega_{i_n}$). Since M is closed, it has a mod-2 fundamental class $[M] \in H_k(M, \mathbb{Z}/2\mathbb{Z})$. Pairing these classes, we get $\langle \omega_{i_1} \cdots \omega_{i_n}, [M] \rangle \in \mathbb{Z}/2\mathbb{Z}$. We will call these numbers the Stiefel-Whitney numbers of M .

Due to proposition 1, Stiefel-Whitney numbers are bordism-invariant. Therefore, we have a homomorphism of Abelian groups $\Omega_k \rightarrow \bigoplus \mathbb{Z}/2\mathbb{Z}$, where the number of copies of $\mathbb{Z}/2\mathbb{Z}$ is equal to the number of partitions of k .

Exercise 5. Prove that this homomorphism is injective. Deduce that Ω_k is isomorphic to some $\bigoplus \mathbb{Z}/2\mathbb{Z}$.

Non-oriented Cobordisms

Proposition 2. (Thom) The non-oriented bordism ring is a polynomial ring $\mathbb{Z}/2\mathbb{Z}[X_2, X_4, X_5, X_6, X_8, \dots]$ with exactly one generator X_n of degree n for any $n \neq 2^t - 1$. Thom proved that $\mathbb{RP}^n$ is a generator only for even values of n . C.T.C Wall then constructed odd-dimensional generators in a later paper.

Proposition 3. (Pontrjagin, Thom, Wall, Averbuh): A closed k -dimensional oriented smooth manifold bounds (i.e. is the boundary of a compact $(k+1)$ -dimensional oriented smooth manifold) if and only if its Stiefel-Whitney numbers and Pontrjagin numbers (existing as cohomology classes of $H^i(M, \mathbb{Z})$ only in $4t$ -dimensional manifolds M) are zero.

Exercise 6. Similar to exercise 5, deduce that $\Omega_k^{\text{Ori.}}$ is isomorphic to some $\bigoplus \mathbb{Z}/2\mathbb{Z} \oplus \bigoplus \mathbb{Z}$.

Proposition 4. (Thom) The oriented bordism ring, modulo the ideal consisting of 2-torsion elements is a polynomial ring $\mathbb{Z}[Y_4, Y_8, Y_{12}, \dots]$ with exactly one generator Y_n of degree n for any $n = 4t$. For $n = 4, 8, 12$, the space $\mathbb{CP}^n$ is a generator. However, $\mathbb{CP}^{16}$ is not a generator. The description of 2-torsions is due to Wall.

Complex Cobordisms

Definition 10. Given a real vector bundle $\pi : \xi \rightarrow M$ of rank $2n$, a **complex structure** is a continuous, π -invariant map $J : \xi \rightarrow \xi$ that is $\mathbb{R}$ -linear on the fibres, and such that $J^2 = -\text{id}_\xi$. This gives the fibres the structure of a $\mathbb{C}$ -vector space given by: $(a + ib).v := av + J(bv)$, $\forall a + ib \in \mathbb{C}$, $v \in \pi^{-1}(p)$, $p \in M$.

Reminder. Given a smooth manifold M , an **almost-complex structure** on M is a complex structure on TM . An **almost-complex manifold** is a manifold equipped with an almost-complex structure.

Definition 11. A smooth manifold M is said to be a **weakly-complex manifold** if $TM \oplus \mathcal{E}^k$ is a complex vector bundle, where $\mathcal{E}^k$ is a trivial bundle.

Exercise 7. Prove that an orientation of TM is equivalent to an orientation of $\mathcal{E}^k$.

Example 3. Let $U \subseteq \mathbb{C}^n$ be an open subset. We then have $TU = U \times \mathbb{C}^n$ and a canonical almost-complex structure on U given by $J_0(u, v) = (u, iv)$. Let $f : U \rightarrow V$ be a smooth map where $V \subseteq \mathbb{C}^n$ is an open subset. Then it is particularly interesting if the differentiation map $df_u : T_u U \rightarrow T_{f(u)} V$ is **complex-linear** with respect to the almost-complex structure on U ; that is, $df \circ J_0 = J_0 \circ df$. Generally, if $f : M \rightarrow N$ is a smooth map of manifolds and J is an almost-complex structure for M , we say that f is **holomorphic with respect to J** if $df \circ J = J \circ df$.

Definition 12. A manifold M equipped with an almost-complex structure J is **complex** if for all $p \in M$, there exists a neighbourhood $\mathcal{U}_p \subseteq M$ and a diffeomorphism $f : \mathcal{U}_p \rightarrow U \subseteq \mathbb{C}^n$ (where $U \subseteq \mathbb{C}^n$ is an open subset) that is holomorphic with respect to J .

Proposition 5. (Milnor, Novikov) A closed k -dimensional weakly-complex manifold bounds (i.e. is the boundary of a compact $(k+1)$ -dimensional weakly-complex manifold) if and only if its Chern numbers are zero.

Proposition 6. (Milnor) The weakly-complex bordism ring is a polynomial ring $\mathbb{Z}[R_2, R_4, \dots]$ with exactly one generator R_n of degree n for any $n = 2t$. As generators one can take certain algebraic varieties with their natural complex structures. It is not known whether connected varieties will suffice.

There are many more variations of the bordism ring, such as ring of symplectic (co)bordisms, measured (co)bordisms and (co)bordisms of spin manifolds. Much less is known about these bordism rings. One thing worthy to note is that to define classical bordism relation, we do not actually need the smooth structure of the manifold. In fact, one can define the oriented and non-oriented bordism rings in the category of topological manifolds and combinatorial

manifolds. It is therefore natural to ask whether these rings are actually different. The canonical forgetful functor from the category of smooth manifolds to the category of topological manifolds induces a homomorphism $\Omega_k^{\text{Sm}} \rightarrow \Omega_k^{\text{Top}}$. Since Thom has shown that Stiefel-Whitney classes can be defined purely topologically, one can immediately see that this homomorphism is injective. It is not known if this homomorphism is also surjective.

In the combinatorial case, we know by a celebrated theorem of Whitehead that any differentiable manifold has a preferred class of triangulations. Therefore, we get a canonical functor from the category of smooth manifolds to the category of combinatorial manifolds. This in turn induces a natural homomorphism $\Omega_k^{\text{Sm}} \rightarrow \Omega_k^{\text{Comb}}$. Thom, Rohlin and Švarc have shown that Pontrjagin classes can be defined for combinatorial manifolds, which immediately proves that this homomorphism is also injective. However, specific examples show that this homomorphism is not surjective.

Another interesting example is the bordism ring of knots. A knot (an embedding of S^1 in S^3) is said to bound if it can be extended to a smooth embedding of D^2 to D^4 . This cobordism ring has been studied by Fox, Milnor and is known not to be finitely-generated unlike the smooth case.

3 A Thorough Study of Complex Cobordisms

One knows very well how to interpret elements of $MU_*(X)$ as bordism classes of maps $f : M \rightarrow X$ where M is a closed weakly-complex manifold. It will be convenient to have a similar geometric picture for cobordism elements. We begin by considering smooth maps of manifolds $f : Z \rightarrow X$, and intend to generalize the concept of a weakly-complex structure on a smooth manifold to maps of smooth manifolds.

Definition 13. Given a smooth map of manifolds $f : Z \rightarrow X$, we define the **relative dimension of f** at $z \in Z$ as $\dim_z(f) = \dim_z(Z) - \dim_{f(z)}(X)$, where $\dim_z(Z)$ is the dimension of Z in the connected component containing z . Suppose that f is of an even relative dimension. Then by a **complex orientation of f** we mean an equivalence class of factorizations of f to $Z \xrightarrow{i} E \xrightarrow{\pi} X$ where $p : E \rightarrow X$ is a complex vector bundle and i is an embedding equipped with a complex structure on its normal bundle. Two complex orientations $Z \xrightarrow{i} E \xrightarrow{\pi} X$ and $Z \xrightarrow{i'} E' \xrightarrow{\pi'} X$ are considered to be equivalent if E, E' can be embedded as subvector-bundles of a vector bundle E'' (therefore i, i' can both be viewed as embeddings into E'') and that there exists an isotopy $i'' : Z \times I \rightarrow E''$ of i, i' whose normal bundle is also equipped with a complex structure which matches the complex structure of the normal bundles of i, i' (namely $N_i, N_{i'}$) at the ends. (note that $N_{i''}|_{Z \times \{0\}} \cong N_i, N_{i''}|_{Z \times \{1\}} \cong N_{i'}$) The definition for f of odd relative dimension is essentially the same, except E is replaced by $E \times \mathbb{R}$. For a general map f , one can write $Z = Z' \sqcup Z''$ where $f' : Z' \rightarrow X, f'' : Z'' \rightarrow X$ are the even-dimensional and odd-dimensional parts of f and define a complex orientation of f as a complex orientation for f', f'' .

Reminder. Given an embedding of manifolds $i : Z \rightarrow E$, the normal bundle is defined as $N_i := TE|_Z / T_Z$.

Definition 14. Two proper, complex-oriented maps $f_0 : Z_0 \rightarrow X, f_1 : Z_1 \rightarrow X$ are **cobordant over X** if there is a smooth manifold W and a proper, complex-oriented map $F : W \rightarrow X \times \mathbb{R}$ such that the following hold:

1. Time slice maps $\epsilon_0, \epsilon_1 : X \rightarrow X \times \mathbb{R}$ defined by $\epsilon_t(x) = (x, t)$ are transversal to F .
2. The smooth submanifolds $F^{-1} \circ \epsilon_0(X), F^{-1} \circ \epsilon_1(X) \subseteq W$ are diffeomorphic to Z_0, Z_1 respectively.
3. The pullback of F by ϵ_0, ϵ_1 have the same complex orientation (pulled back from F) as f_0, f_1 respectively.

Exercise 8. Cobordism over X is an equivalence relation on the space of proper complex-oriented maps $Z \rightarrow X$.

Proposition 7. For a smooth manifold X , the set of cobordism classes of proper, complex-oriented maps $Z \rightarrow X$ of relative codimension q is canonically isomorphic to the cohomology group $MU^q(X)$.

Exercise 9. Translate and simplify this theorem in the case where $X = \text{pt}$, which is Thom's seminal theorem.

We now study the properties of cobordisms using the provided geometric machinery.

Property 1. (Contravariant variance: First Gysin homomorphism) Let $g : Y \rightarrow X$ be a smooth map and $f : Z \rightarrow X$ be a proper, complex-oriented map. By Thom's transversality theorem, one can assume, without changing the homotopy class of g , that $g \pitchfork f$. Therefore, the pullback $g^*f : Y \times_X Z \rightarrow Y$ exists and is a proper map which inherits the complex orientation of f . Furthermore, $\dim Y \times_X Z = \dim Y + \dim Z - \dim X$. One can easily check that pullback of cobordant maps over X are cobordant over Y and that pullback preserves the relative codimension. Therefore, we get a map $g^* : MU^q(X) \rightarrow MU^q(Y)$, $\forall q$.

Property 2. (Covariant variance: Second Gysin homomorphism) Let $g : X \rightarrow Y$ be a proper, complex-oriented map of relative codimension d . Then we can assign to each proper, complex-oriented map $f : Z \rightarrow X$ in $MU^q(X)$, the composition $g \circ f : Z \rightarrow Y$ which is a proper, complex-oriented map of codimension $q + d$. One can see that this composition preserves cobordisms and therefore, we get a map $g_* : MU^q(X) \rightarrow MU^{q+d}(Y)$, $\forall q$.

Property 3. (Additive structure) We define the sum of cobordism classes of maps $f_0 : Z_0 \rightarrow X$, $f_1 : Z_1 \rightarrow X$ as the cobordism class of the map $f_0 \sqcup f_1 : Z_0 \sqcup Z_1 \rightarrow X$ and denote it by $[f_0] + [f_1]$. The identity element of $MU^q(X)$ is the vacuous map $\emptyset \rightarrow X$. The negative of the cobordism class of $f : Z \rightarrow X$ is the cobordism class of f endowed with **negative complex orientation**. If f is of even relative dimension, its complex orientation is represented by a factorization $Z \xrightarrow{i} E \xrightarrow{\pi} X$. One can now locally trivialize E to get $E|_U \cong U \times \mathbb{C}^n$. Since the complex structure J on E is π -invariant, this induces a complex structure on $E|_U$; i.e. a continuous π -invariant map $J|_U : U \times \mathbb{C}^n \rightarrow U \times \mathbb{C}^n$ that is $\mathbb{R}$ -linear on the fibres. Now simply change the complex structure to $-J$. The construction for the odd relative dimension and general maps are similar.

Property 4. (Product structure) The product of cobordism classes of maps $f_0 : Z_0 \rightarrow X$, $f_1 : Z_1 \rightarrow X$ is defined as the cobordism class of the map $f_0 \times f_1 : Z_0 \times Z_1 \rightarrow X_0 \times X_1$ and denoted by $[f_0] \otimes [f_1] \in MU^*(X_1 \times X^2)$. One can show that this product is well-defined. Furthermore, it induces a ring structure on $MU^*(X)$ given by $[f_0] \cdot [f_1] = \Delta^*([f_0] \otimes [f_1])$ where $\Delta : X \rightarrow X \times X$ is the diagonal morphism.

The interested reader is encouraged to prove that the additive and product structure are well-defined. We now show that the Gysin homomorphisms characterize MU^* as a functor from the category **Diff** of smooth manifolds to **Sets**.